



CPIT240 | Lab Exam - (10 marks)

Spring 2024

Instructions:

- Download the exam pdf ([/Quizzes/Tests/Lab Exam](#)) and read the requirement carefully.
- Use Oracle Live SQL to answer the questions <https://livesql.oracle.com/apex/f?p=590:1000>
- After answering all questions, save your session as a script file named **YourName_ID_Section_LabExam** (please follow the saving steps demonstrated in the file named *Saving Your Session as Script in Oracle Live SQL* on Blackboard)
- Edit your answers carefully before submitting them to the blackboard.

Submission Guidelines:

Submit to blackboard your script file (**YourName_ID_Section_LabExam.sql**) attached with the [share link](#) (step 3 in the file named *Saving Your Session as Script in Oracle Live SQL* on Blackboard).

Part-1 Tasks: [5 Marks]

Write the SQL statements that perform the following:

1. Create **CUSTOMERS** and **ORDERS** tables based on the following table instance charts. [2 Marks]

Customers Table Instance Chart

Column Name	Cust_ID	Cust_name	Cust_address
Constraint	PK		
Data Type	NUMBER	VARCHAR2	VARCHAR2
Length	8	20	30

Orders Table Instance Chart

Column Name	O_ID	O_date	Cust_ID
Constraint	PK	NOT NULL	FK
Data type	NUMBER	DATE	NUMBER
Length	5		8

2. Add a **NOT NULL** constraint to the **Cust_name** column in the CUSTOMERS table. [1 Mark]
3. Insert the **CUSTOMERS** and **ORDERS** information. [0.5 Mark]

Customers Information

Cust_ID	Cust_name	Cust_address
1	Ahmad	Jeddah
2	Salman	Riyadh
3	Saleh	Abha

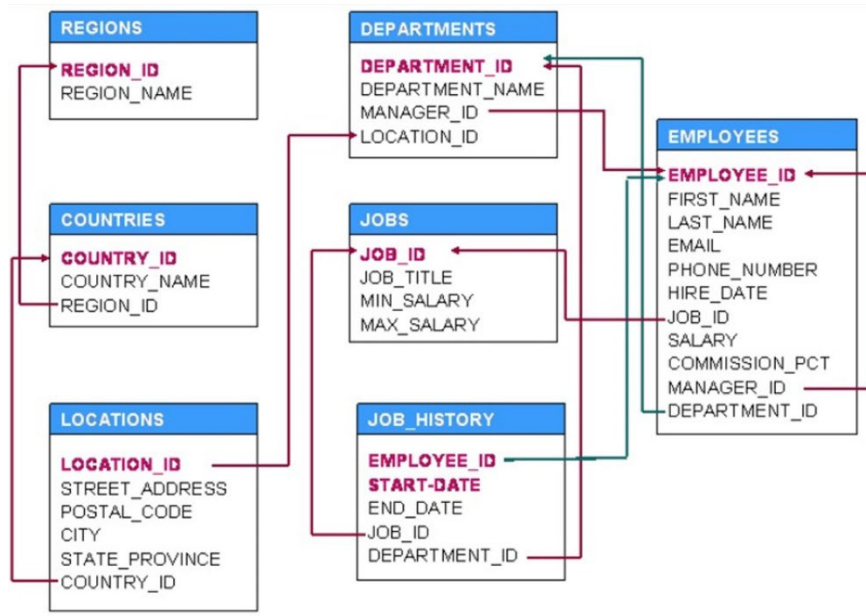
Orders Information

O_ID	O_date	Cust_ID
11	4-MAY-21	2
22	3-JUN-23	3
33	2-AUG-24	3
44	1-MAY-20	1

4. Display the **name** of all customers in descending order. [0.5 Mark]
5. Delete the orders that were ordered in **MAY**, no matter the day and the year. [1 Mark]

Part-2 Tasks: [5 Marks]

Use the HR schema from Oracle Live SQL to answer the following questions, below is the relational model for the HR schema.



The required tables for the following tasks are (**employees, departments, and locations**).

6. Retrieve each department name with Its manager's first name. The result should have only one column called "Managers" and each tuple should be in the form "**Sarah manages the Shipping department**". [1 Mark]
7. Use the employees table to show the **department_id** and the **average** salary for each department. [1 Mark]
8. Retrieve the **department ID, department name, and manager_id** for all departments that have **no** manager. [1 Mark]
9. Retrieve the **employee ID, first name, and the job title** of all employees (*Note: use NATURAL JOIN*). [1 Mark]
10. Use the employees table to show **the employee ID, first name, and the department ID** for all employees who works in the same department as employee **115**. [1 Mark]